Reading notes: arXiv:2306.01039

Paper. K. Budzik, D. Gaiotto, J. Kulp, B. R. Williams, J. Wu, M. Yu, *Semi-Chiral Operators in 4d $\mathcal{N}=1$ Gauge Theories*, arXiv:2306.01039 [hep-th] (id verified on the abs page and on the PDF title page); v1 1 Jun 2023, v2 3 Sep 2024 (= published version), JHEP 05 (2024) 245, DOI 10.1007/JHEP05(2024)245. 55+20 pages.

Source used. PDF of v2 (arxiv.org/pdf/2306.01039v2, read page by page). Equation numbers below are those of the v2 PDF (section-numbered, e.g. (3.40)). The arXiv HTML rendering of v2 numbers equations consecutively and splits multi-line displays; key correspondences: (3.24)=HTML(55), (3.27)=(58), (3.28)=(59), (3.30)=(61), (3.33)=(64), (3.36)=(67), (3.39)=(70), (3.40)=(71), (3.41)–(3.45)=(72)–(76), (4.1)=(77), (5.20)=(128), (5.21)=(129). Page numbers "p. n" are the printed page numbers (PDF page = $n+2$).

Name used in these notes. "Q_1" always means the paper's one-loop term of the loop-corrected twisted differential, (3.28), (D.13). It is *not* the spinor parameter $Q_1 \in \Sigma$ of the manuscript (Prop. prop:c7-susy). Prose of the paper is paraphrased; displayed formulas are transcribed.

1 Conventions of the paper

Signature, spinors. Euclidean $\mathcal{N}=1$ algebra; Spin(4) $\cong SU(2)\times SU(2)$ (p. 5). Footnote 7: the paper *switches* the standard (Wess–Bagger [34]) assignment of dotted/undotted indices.

$$\{Q_{\dot\alpha}, \bar Q_\beta\} = P_{\dot\alpha\beta}, \tag{2.1}$$
$$\{Q_{\dot\alpha}, Q_{\dot\beta}\} = \{\bar Q_\alpha, \bar Q_\beta\} = 0. \tag{2.2}$$

Rotations: $M_{\dot\alpha\dot\beta}$ and $\bar M_{\alpha\beta}$ (p. 5). No normalisation of $P_{\dot\alpha\beta}$ in terms of σ-matrices is given.

Twisting supercharge and complex structure (p. 6). $Q \equiv Q_{\dot-}$; $z^\alpha \equiv x^{\dot+\alpha}$ holomorphic, $\bar z^\alpha \equiv x^{\dot-\alpha}$ antiholomorphic; footnote 9: $(z_\alpha)^* = \epsilon_{\alpha\beta}\bar z^\beta$.

$$\{Q, \bar Q_\alpha\} = \partial_{\dot-\alpha} = \partial_{\bar z^\alpha}. \tag{2.3}$$

Surviving rotations: the $SU(2)$ generated by $\bar M_{\alpha\beta}$ (holomorphic symplectic rotations, preserving $\mathrm{d}^2 z = \mathrm{d}z^1\wedge\mathrm{d}z^2$). With $U(1)_R$: $M_R \equiv M_{\dot+\dot-} - R$ extends it to $U(2)$. Footnote 10: $[M_{\dot+\dot-}, Q_{\dot-}] = -Q_{\dot-}$, $[R, Q_{\dot-}] = -Q_{\dot-}$. Cohomological degree $C \equiv \mathrm{gh} - M_{\dot+\dot-}$.

Charges (Table 1, p. 50):

	R	$M_{\dot+\dot-}$	M_R	C_R	C
Q	-1	-1	0	1	1
b	0	2	2	0	-2
c	-1	-1	0	1	1
∂_α	0	1	1	0	-1

$$M_R = M_{\dot+\dot-} - R, \quad C_R = \mathrm{gh} - R, \quad C = C_R - M_R = \mathrm{gh} - M_{\dot+\dot-}.$$

Superconformal relations (for SCFTs, kinematics only):

$$\{Q_{\dot-}, S_{\dot+}\} = \Delta + \tfrac{3}{2}R - M_{\dot+\dot-}, \tag{2.4}$$
$$\{Q_{\dot\alpha}, S_{\dot\beta}\} = \big(\Delta + \tfrac{3}{2}R\big)\epsilon_{\dot\alpha\dot\beta} - M_{\dot\alpha\dot\beta}. \tag{2.5}$$

Short $\mathfrak{su}(1|1)$ multiplets: $\Delta = M_{\dot+\dot-} - \frac{3}{2}R$ (p. 7). Statement (p. 7): in SCFTs Q-cohomology classes are annihilated by $M_{\dot+\dot+}$ and have non-positive cohomological degree; degree 0 = chiral ring.

Chiral superspace and descent (p. 8–9). $\bar{Q}_\alpha = -\partial_{\bar{\theta}^\alpha}$ on superfields; $\bar{\theta}^\alpha \leftrightarrow \mathrm{d}\bar{z}^\alpha$.

$$\bar{\partial} \equiv \bar{\theta}^\alpha \partial_{\bar{z}^\alpha}, \tag{2.6}$$

$$\mathcal{O}[\bar{\theta}] = e^{\bar{\theta}^\alpha \bar{Q}_\alpha} \mathcal{O}^{(0)} = \mathcal{O}^{(0)} + \mathcal{O}^{(1)} + \mathcal{O}^{(2)}, \tag{2.7}$$

$$\{Q + \bar{\partial}, \bar{Q}_\alpha\} = 0, \tag{2.8}$$

$$(Q + \bar{\partial})\mathcal{O}[\bar{\theta}] = (Q\mathcal{O})[\bar{\theta}], \tag{2.9}$$

$$Q\mathcal{O}^{(k)} + \bar{\partial}\mathcal{O}^{(k-1)} = (Q\mathcal{O})^{(k)}, \tag{2.10}$$

$$D_{\dot{-}}\mathcal{O} = (Q + \bar{\partial})\mathcal{O} = 0 \quad \text{(semi-chiral)}, \tag{2.11}$$

$$Q\mathcal{O}^{(1)} + \bar{\partial}\mathcal{O}^{(0)} = 0, \qquad Q\mathcal{O}^{(2)} + \bar{\partial}\mathcal{O}^{(1)} = 0, \tag{2.12}$$

$$\mathcal{O} = (Q + \bar{\partial})\mathcal{U} \quad \text{if } \mathcal{O}^{(0)} = Q\mathcal{U}^{(0)}. \tag{2.13}$$

$Q + \bar{\partial}$ coincides with the superspace derivative $D_{\dot{-}}$ of the physical theory (p. 8).

λ-bracket (n complex dimensions):

$$\{\mathcal{O}_1\,{}_\lambda\,\mathcal{O}_2\} = \oint_{S^{2n-1}} e^{\lambda\cdot z}\,\mathrm{d}^n z\,\mathcal{O}_1(z)\mathcal{O}_2(0). \tag{2.18}$$

Lie algebra (p. 19–20). Killing form δ^{AB}, adjoint indices raised/lowered freely; trace in the defining representation;

$$\mathrm{Tr}\, t^A t^B = -\tfrac{1}{2}\delta^{AB}, \qquad \mathrm{Tr}\, t^A [t^B, t^C] = -\tfrac{1}{2} f^{ABC}, \qquad [t^A, t^B] = f^{ABC} t^C, \tag{3.20}$$

$$\textstyle\sum_{A,B} f^{ABC} f^{ABD} = N\delta^{CD}, \tag{3.21}$$

$$SU(2):\ t^A = -\tfrac{i}{2}\sigma^A,\ f^{ABC} = \epsilon^{ABC}; \qquad Qb^A = \epsilon^{ABC} c^B b^C, \quad Qc^A = \tfrac{1}{2}\epsilon^{ABC} c^B c^C, \tag{3.22}$$

$$b^2 = b_A b^A = -2\,\mathrm{Tr}\, b^2. \tag{3.23}$$

Tree bracket (App. B, p. 57).

$$\{b^A, c^B\} = \delta^{AB}, \tag{B.1}$$

$$\{\mathcal{O}_1, \mathcal{O}_2\} = \delta^{AB} \frac{\delta\mathcal{O}_1}{\delta c^A}\frac{\delta\mathcal{O}_2}{\delta b^B} + (-1)^{|\mathcal{O}_1|}\delta^{AB}\frac{\delta\mathcal{O}_1}{\delta b^A}\frac{\delta\mathcal{O}_2}{\delta c^B}, \tag{B.2}$$

$$\frac{\delta}{\delta c^A}\big(c^B c^C\big) = \delta^{AB} c^C - \delta^{AC} c^B. \tag{B.3}$$

Footnote 54 (App. B.2): $\psi^\alpha \chi_\alpha = \psi_2\chi_1 - \psi_1\chi_2 = -\psi_\alpha \chi^\alpha$.

Coupling and $\hbar$. The BF action (3.2) contains no coupling; τ appears only in (3.4), a total derivative in BF theory (instantons only). $\hbar$ appears in (3.27) and in (D.13): $Q = Q_0 + \hbar Q_1 + \hbar^2 Q_2 + \cdots$; (3.28) writes $\mathbf{Q} = Q_0 + Q_1 + \cdots$ with Q_n the n-loop term. No relation between $\hbar$ and the physical g^2 (or $g^2 N/16\pi^2$) is given.

2 Twisted theory and its relation to the untwisted fields

Pure $\mathcal{N}=1$ gauge theory = holomorphic BF theory (p. 13–15). The identification (in BV, refs. [22,41,50]) changes the field content but not the Q-cohomology. $Y=\mathbb{C}^2$.

$$b=b^{(0)}+b^{(1)}+b^{(2)}\in\Omega^{0,\bullet}(\mathbb{C}^2,\mathfrak{g}^\vee),\qquad c=c^{(0)}+c^{(1)}+c^{(2)}\in\Omega^{0,\bullet}(\mathbb{C}^2,\mathfrak{g})[1], \tag{3.1}$$

$$\int \mathrm{d}^2z\ \mathrm{Tr}\ b\Big(\bar\partial c-\tfrac12[c,c]\Big), \tag{3.2}$$

$$\int \mathrm{d}^2z\ \mathrm{Tr}\ b^{(0)}\Big(\bar\partial c^{(1)}-\tfrac12[c^{(1)},c^{(1)}]\Big), \tag{3.3}$$

$$\int \mathrm{d}^2z\,\tau\ \mathrm{Tr}\ \partial_\alpha c\,\partial^\alpha c. \tag{3.4}$$

Degrees: $c^{(0)}$ ghost, degree 1; $c^{(1)}$ degree 0 = antiholomorphic part of the connection; $b^{(0)}$ degree -2; the holomorphic part of the connection is absent; the 2-form part of $\bar\partial c-\frac12[c,c]$ is the antiholomorphic field strength. Footnote 21: $b^{(i)}$ is naturally a $(2,i)$ form; d^2z fixed, so $(0,i)$. Footnote 22: c adjoint, b coadjoint, identified by an invariant pairing.

Dictionary to the physical fields (p. 14, stated up to field redefinitions, at the level of Q-cohomology of the free theory):

- $(2,0)$ part of $F_{++}\leftrightarrow \mathrm{d}^2z\,b^{(0)}$;
- $\bar\lambda_\alpha$ ($\alpha=1,2$), with $\epsilon_{\alpha\beta}\partial^\alpha\bar\lambda^\beta=\partial_\alpha\bar\lambda^\alpha=0\leftrightarrow\partial_{z^\alpha}c^{(0)}$; superfield: $\overline{\mathcal{W}}_\alpha$ (anti-chiral) $\leftrightarrow\partial_{z^\alpha}c$;
- kinetic F-term $\int\mathrm{d}^2\bar\theta\,\tau\,\overline{\mathcal{W}}_\alpha\overline{\mathcal{W}}^\alpha\leftrightarrow$ (3.4).

The basic action corresponds to a self-dual limit of the twisted theory (p. 14). Footnote 20: on a curved complex surface Y the one-loop quantisation has an anomaly $\propto c_1(T_Y)\times$ adjoint quadratic Casimir.

Chiral multiplet (§2.5):

$$\gamma\equiv\Phi, \tag{2.23}$$

$$\beta\equiv D_+\bar\Phi, \tag{2.24}$$

$$S_\alpha=\beta\partial_\alpha\gamma-\Delta\,\partial_\alpha(\beta\gamma), \tag{2.25}$$

$$\partial_\alpha S^\alpha=\partial_\alpha\beta\,\partial^\alpha\gamma, \tag{2.26}$$

$$(Q+\bar\partial)\beta=\partial_\gamma\mathcal{W}(\gamma), \tag{2.27}$$

$$(Q+\bar\partial)\,\beta\partial_\alpha\gamma=\partial_\alpha\big[\mathcal{W}-\Delta\,\gamma\partial_\gamma\mathcal{W}\big]. \tag{2.28}$$

$J=\beta\gamma$ from $|\Phi|^2$; Δ in (2.25) $\propto$ R-charge of Φ.

Matter, tree level (§4.1). Chirals in R: bosonic γ in R, fermionic β in $\bar R\simeq R^*$.

$$\int\mathrm{d}^2z\,\beta\big(\bar\partial\gamma-[c,\gamma]\big)+W(\gamma), \tag{4.1}$$

$$Q_0b=-\bar\partial b+[c,b]+\gamma\beta,\qquad Q_0\gamma=-\bar\partial\gamma+[c,\gamma],\qquad Q_0\beta=-\bar\partial\beta+[c,\beta]+\partial_\gamma W, \tag{4.2}$$

$$S_\alpha=\mathrm{Tr}\,b\partial_\alpha c+c_1\tilde\Gamma^A\partial\,\Gamma_A+c_2\,\partial\tilde\Gamma^A\Gamma_A\quad(\text{SQCD, constants }c_1,c_2). \tag{4.3}$$

With matter: $\mathrm{Tr}\,b^2$ is not tree-level Q-closed; in the SQCD discussion the stress tensor and its derivatives are stated not to be exact, at least perturbatively (p. 28). Chiral operators $\leftrightarrow$ G-invariant polynomials in γ modulo multiples of $\partial_\gamma W$.

$\mathcal{N}=4$ **SYM** (p. 28, 31). Twist = holomorphic bc system in $\mathfrak{gl}(N)$ + three adjoint $\beta\gamma$ systems (β_i,γ^i), $i=1,2,3$; superpotential $\mathrm{Tr}[\gamma^1,\gamma^2]\gamma^3$ (p. 28), written $W=\mathrm{Tr}\,\gamma^1[\gamma^2,\gamma^3]$ on p. 31 (equal by cyclicity). Following [19]:

$$C(\varepsilon_1,\varepsilon_2,\varepsilon_3)\equiv c+\varepsilon_i\gamma^i-\tfrac{1}{2}\epsilon^{ijk}\varepsilon_i\varepsilon_j\beta_k-\varepsilon_1\varepsilon_2\varepsilon_3\,b, \tag{4.4}$$

$$Q_0C(\varepsilon_i)=\tfrac{1}{2}[C(\varepsilon_i),C(\varepsilon_i)], \tag{4.5}$$

$$C^\bullet\big(\mathfrak{gl}_N\otimes\mathbb{C}[[z_1,z_2,\varepsilon_i]]\,|\,\mathfrak{gl}_N\big), \tag{4.6}$$

$$Q_0c=\tfrac{1}{2}[c,c],\quad Q_0b=[c,b]+\sum_{i=1}^{3}[\beta_i,\gamma^i],\quad Q_0\gamma^i=[c,\gamma^i],\quad Q_0\beta_i=[c,\beta_i]+\tfrac{1}{2}\sum_{j,k=1}^{3}\epsilon_{ijk}[\gamma^j,\gamma^k]. \tag{4.14}$$

Enhanced tree-level symmetry: holomorphic vector fields on $\mathbb{C}^{2|3}$ (coordinates z_α,ε_i) preserving $\mathrm{d}^2z\,\mathrm{d}^3\varepsilon$ (p. 29). *No one-loop formula for $\mathcal{N}=4$ is given in this paper.*

3 Local operators: free and tree level

$$\mathcal{V}\equiv\mathbb{C}\Big[\partial_{z^1}^n\partial_{z^2}^m c^{(0)},\ \partial_{z^1}^k\partial_{z^2}^l b^{(0)},\ n+m>0,\ k+l\geq 0\Big]^G, \tag{3.5}$$

$$Qc=-\bar\partial c+\tfrac{1}{2}[c,c],\qquad Qb=-\bar\partial b+[c,b], \tag{3.6}$$

$$(Q+\bar\partial)\partial_\alpha c=[c,\partial_\alpha c],\quad (Q+\bar\partial)\partial_\alpha\partial_\beta c=[c,\partial_\alpha\partial_\beta c]+[\partial_\alpha c,\partial_\beta c],\quad (Q+\bar\partial)\partial_\alpha b=[c,\partial_\alpha b]+[\partial_\alpha c,b], \tag{3.7}$$

$$Q=Q_{\mathrm{free}}+Q_{\mathrm{int}},\qquad Q_{\mathrm{free}}=-\bar\partial, \tag{3.8}$$

$$Q_{\mathrm{int}}c=\tfrac{1}{2}[c,c],\qquad Q_{\mathrm{int}}b=[c,b]. \tag{3.9}$$

$c^{(0)}$ itself is excluded from local operators (only derivatives; constant gauge transformations imposed as strict invariance, p. 15). Free theory: kinetic term pairs $b^{(0)}$ with $c^{(1)}$ and $b^{(1)}$ with $c^{(0)}$, so $b^{(0)},c^{(0)}$ have regular OPE and their composites have no renormalisation ambiguity in the free theory; Δ_{BV} must be regularised (footnote 24); quantum corrections do not change $\mathcal{V}$ in the free theory (p. 15–16). Tree level: Q_{int} reproduces the categorification literature [18,19,51], with Q^2_{-} acting on D_{z^α} as $[\bar\lambda_\alpha,\cdot\,]$ (p. 17).

$$C^\bullet\big(\mathfrak{g}[[z_1,z_2]]\,|\,\mathfrak{g};\ \mathrm{Sym}(\mathfrak{g}[[z_1,z_2]]^*)\big), \tag{3.10}$$

$$A_n=\mathrm{Tr}\,b^n,\ n\geq 1, \tag{3.11}$$

$$C(\varepsilon)\equiv c+\varepsilon b,\qquad Q_{\mathrm{int}}C(\varepsilon)=\tfrac{1}{2}[C(\varepsilon),C(\varepsilon)], \tag{3.12}$$

$$\xi=\partial_\varepsilon:\ c\to b,\qquad \tilde\xi_\alpha=\varepsilon\partial_\alpha:\ b\to\partial_\alpha c, \tag{3.14}$$

$$B_{a-1,\alpha}\equiv\tilde\xi_\alpha A_a,\qquad C_{a-2}\equiv\tilde\xi_1\tilde\xi_2 A_a, \tag{3.15}$$

$$\xi B_{a-1,\alpha}=\partial_\alpha A_a,\qquad \xi C_{a-2}=2\partial_\alpha B^\alpha_{a-1}, \tag{3.16}$$

$$\partial_\alpha B^\alpha_{a-1}=\tilde\xi^\alpha\partial_\alpha A_a, \tag{3.17}$$

$$A_n=\mathrm{Tr}\,b^n\ (n\geq 2),\qquad B_{n,\alpha}=\mathrm{STr}\,b^n\partial_\alpha c\ (n\geq 1),\qquad C_n=\mathrm{STr}\,b^n\partial_\alpha c\,\partial^\alpha c\ (n\geq 0), \tag{3.18}$$

$$S_\alpha=\mathrm{Tr}\,b\,\partial_\alpha c. \tag{3.19}$$

ξ generated by $A_2=\mathrm{Tr}\,b^2$; $\tilde\xi_\alpha$ by $\mathrm{Tr}\,c\,\partial_\alpha c$ (primitive of C_0) (p. 19). Large N (§5): $\mathrm{Tr}[\mathrm{d}C]^n$ with $\mathrm{d}C=\mathrm{d}\varepsilon\,b+\mathrm{d}z_\alpha\partial_\alpha c+\varepsilon\,\mathrm{d}z_\alpha\partial_\alpha b$ (5.8);

$$\begin{aligned}\mathrm{Tr}[\mathrm{d}C]^n&=\mathrm{d}\varepsilon^n A_n+\mathrm{d}z_\alpha\mathrm{d}\varepsilon^{n-1}B_{n-1,\alpha}+\mathrm{d}z^1\mathrm{d}z^2\mathrm{d}\varepsilon^{n-2}C_{n-2}\\&\quad+\varepsilon\,\mathrm{d}z_\alpha\mathrm{d}\varepsilon^{n-1}n^{-1}\partial_\alpha A_n+\varepsilon\,\mathrm{d}z^1\mathrm{d}z^2\mathrm{d}\varepsilon^{n-2}(n-1)^{-1}\partial_\alpha B_{n-1,\alpha},\end{aligned} \tag{5.10}$$

$$A_n=\mathrm{Tr}\,b^n\ (n\geq 1),\quad B_{n,\alpha}=\mathrm{Tr}\,b^n\partial_\alpha c\ (n\geq 1),\quad C_n=\mathrm{Tr}\,b^n\partial_\alpha c\,\partial^\alpha c\ (n\geq 0). \tag{5.19}$$

Large-N single-trace tree cohomology: (5.19) with derivatives, plus derivatives of $\operatorname{Tr} c$ (p. 41). B.2 normalisation (different): $A_n = \frac{(2i)^n}{n} \operatorname{Tr} b^n$, $A_n[b + \eta^\alpha \partial_\alpha c] = A_n + \eta^\alpha B_{n-1,\alpha} + \eta^1\eta^2 C_{n-2}$ (B.12); tree λ-brackets (B.13) hold up to Q_0-exact terms. App. B.1: stress tensor of the bc system written $2 \operatorname{Tr} b\, \partial_\alpha c$ (factor 2 relative to (3.19)), with

$$g_{m,n} = z_1^{m+1} z_2^{n+1} \{\partial_\alpha S^\alpha, -\}, \qquad \partial_\alpha S^\alpha = 2 \operatorname{Tr} \partial_\alpha b\, \partial^\alpha c = -\partial_\alpha b_A \partial^\alpha c^A, \tag{B.8}$$

$$g_{m,n}\mathcal{O} = -X_{f_{m,n}}\mathcal{O}, \qquad X_f = \frac{\partial f}{\partial z_1}\frac{\partial}{\partial z_2} - \frac{\partial f}{\partial z_2}\frac{\partial}{\partial z_1}, \tag{B.10, B.4}$$

$$g_{-1,0}\mathcal{O} = \partial_1 \mathcal{O}, \quad g_{0,-1}\mathcal{O} = -\partial_2 \mathcal{O}, \quad -\tfrac{1}{2}(g_{1,-1} + g_{-1,1})\mathcal{O} = (x_1\partial_2 - x_2\partial_1)\mathcal{O}. \tag{B.11}$$

4 Definition of the loop-corrected differential (what is defined)

$$\bar\partial + \hbar\Delta_{\mathrm{BV}} + \{\mathcal{I}, \cdot\}_{\mathrm{BV}}, \tag{3.27}$$

$$\mathbf{Q} = Q_0 + Q_1 + \cdots . \tag{3.28}$$

(3.27): full quantum differential on operators; proper definition needs renormalisation of UV and IR divergences (p. 21). Conjecture (p. 21): the cohomology of (3.27) is computed by a differential $\mathbf{Q}$ on $\mathcal{V}$, a quantum-corrected Q_{int}. Operational definition (p. 21–22): regularise $\mathcal{O}\, e^{\int \mathcal{I}}$, act with Q_{free}, rewrite the result as the regularisation of $(\mathbf{Q}\mathcal{O})\, e^{\int \mathcal{I}}$. Q_n: n-loop diagrams of the holomorphic theory. Footnote 29: cohomology of the formal series via $Q_0^2 = 0$ and homotopy transfer (App. D).

$$\int_{\mathbb{R}^4} \mathcal{I}^{(2)}(z,\bar z)\, \mathrm{d}^2 z, \qquad \mathcal{I} = -\tfrac{1}{2} \operatorname{Tr} b[c,c], \tag{3.29}$$

$$Q_0\mathcal{O} = Q_{\mathrm{free}}\Big[\int_{|z|>\epsilon} \mathcal{I}(z,\bar z)\mathrm{d}^2 z\ \mathcal{O}(0,0)\Big] = \int_{|z|=\epsilon} \mathcal{I}(z,\bar z)\mathrm{d}^2 z\, \mathcal{O}(0,0), \tag{3.30}$$

$$Q_0\mathcal{O} = \{\mathcal{I}, \mathcal{O}\} \quad (\lambda = 0 \text{ specialisation of (2.18)}), \tag{3.31}$$

$$\int_{|z|=\epsilon} \mathrm{d}^2 z\, F(z,\bar z) P(z,\bar z) \to F(0,0) + O(\epsilon), \tag{3.32}$$

$$Q_n\mathcal{O} = \frac{1}{(n+1)!}\{\mathcal{I}, \mathcal{I}, \cdots, \mathcal{I}, \mathcal{O}\}, \tag{3.33}$$

$$\tfrac{1}{2}\{\mathcal{I},\mathcal{I}\} + \tfrac{1}{6}\{\mathcal{I},\mathcal{I},\mathcal{I}\} + \cdots = 0, \tag{3.34}$$

$$\int_{\mathbb{R}^{4n}} \prod_{i=1}^{n} \mathcal{I}(z_i,\bar z_i)\mathrm{d}^2 z_i\, \mathcal{O}(0,0), \tag{3.35}$$

$$P(x,\bar x,\mathrm{d}\bar x) \equiv \frac{1}{4\pi^2}\frac{\bar x^2 \mathrm{d}\bar x^1 - \bar x^1 \mathrm{d}\bar x^2}{|x|^4}, \qquad |x|^2 = x^1\bar x^1 + x^2 \bar x^2, \tag{3.36}$$

$$\bar\partial P(x,\bar x,\mathrm{d}\bar x) = \mathrm{d}\bar x^1 \mathrm{d}\bar x^2\, \delta^4(x), \qquad \bar\partial = \mathrm{d}\bar x^1 \tfrac{\partial}{\partial \bar x^1} + \mathrm{d}\bar x^2 \tfrac{\partial}{\partial \bar x^2}. \tag{3.37}$$

(3.30): regulator for Q_0 = interaction switched off in a ball of radius ϵ around $\mathcal{O}$ (BRST anomaly at the boundary); as an element of Q_{free}-cohomology it is ϵ-independent. (3.38): each propagator $P(x-y)$ contracts $c(x)$ with $b(y)$ and removes one b and one c. Each $\mathcal{I}$ adds two c and one b. Q_n: integral over $2n+2$ antiholomorphic variables, hence $2n+1$ propagators; net number of c increases by 1, of b decreases by n; each type of derivative increases by n (p. 24). Higher brackets: quadratic axioms + MC condition (3.34) $\Rightarrow$ $\mathbf{Q}$ nilpotent order by order (argued; details in [26]).

Procedure (p. 23–24), paraphrased: (1) integral (3.35); (2) maximal number of Wick contractions with *regularised* propagators; (3) act with Q_{free}, integrate $\bar\partial$ by parts onto the propagators; (4) set descendants to zero, project to the origin ($\bar z_i = 0$, Taylor in z_i) to get an element of $\mathcal{V}$; (5) the resulting

Feynman integrals are claimed finite and regulator independent. The regulator of the propagators is not specified in this paper.

Tree level (p. 24–25): single internal line; Leibniz rule; commutes with ∂_α;

$$Q_0 b(x) = \{\mathcal{I}, b\}(x) = \int_{|y-x|>\epsilon} \mathrm{d}^2 y\, \bar\partial_y P(y-x)[c(y), b(y)] = [c(x), b(x)] - \bar\partial_x \int_{|y-x|>\epsilon} \mathrm{d}^2 y\, P(y-x)[c(y), b(y)], \tag{3.39}$$

i.e. $[c, b]$ up to a Q_{free}-exact term; $Q_0 c = \{\mathcal{I}, c\} = \frac{1}{2}[c, c]$; $Q_0 = Q_{\text{int}}$.

Nilpotency and filtration (p. 26–27):

$$\mathbf{Q}^2 \mathcal{O} = 0 \quad \forall \mathcal{O} \in \mathcal{V}, \tag{3.41}$$

$$\sum_{n_1+n_2=n} Q_{n_1} Q_{n_2} \mathcal{O} = 0 \quad \forall \mathcal{O} \in \mathcal{V},\ \forall n \geq 0, \tag{3.42}$$

$$Q_n V(N_b, N_c, J_1, J_2) = 0 \quad \text{if } N_b + N_c < n + 1, \tag{3.43}$$

$$Q_n : V(n+1, 0, J_1, J_2) \to V(1, 1, J_1 + n, J_2 + n), \tag{3.44}$$

$$Q_n : V(n, 1, J_1, J_2) \to V(0, 2, J_1 + n, J_2 + n). \tag{3.45}$$

$V(N_b, N_c, J_1, J_2)$: monomials with N_b letters b, N_c letters c (with derivatives), J_i derivatives in z_i. Q_n linear; derivation on monomials longer than $n + 1$. First non-trivial equation of (3.42): $\{Q_0, Q_1\} = 0$. $Q_1^2 \neq 0$ (observed), hence $Q_2 \neq 0$; [32] computes the only two-loop diagram, non-vanishing. Auxiliary $U(1)_b$ (b charge 1, c charge 0): Q_0 preserves it, Q_n lowers it; the filtration makes $\mathbf{Q}$ well defined on monomials and the spectral sequence well posed.

Homotopy transfer (App. D.2):

$$Q = Q_0 + \hbar Q_1 + \hbar^2 Q_2 + \cdots, \tag{D.13}$$

$$\Delta \equiv \iota_0 \pi_0 + \{h_0, Q\} = 1_X + \hbar\{h_0, Q_1\} + \hbar^2\{h_0, Q_2\} + \cdots, \tag{D.14}$$

$$\tilde{Q} = \pi_0(\hbar Q_1 + \cdots)\Delta^{-1} i_0 = \hbar Q_0^{(1)} + \hbar^2 Q_1^{(1)} + \cdots. \tag{D.15}$$

Iterating gives a limiting $U_0^{(\infty)}$, taken as the definition of perturbative cohomology (requires a filtration).

5 One-loop formulas (what is computed)

Structure (§3.2.3, App. C).

$$Q_1 \mathcal{O} \equiv \tfrac{1}{2}\{\mathcal{I}, \mathcal{I}, \mathcal{O}\} \quad (\text{all } \lambda = 0). \tag{C.1}$$

Only triangle diagrams: two fields of $\mathcal{O}$ and two fields of each of the two vertices $\mathcal{I}(x), \mathcal{I}(y)$ contracted. On a single field the triple bracket vanishes identically. On longer monomials, sum over pairs:

$$Q_1\big(f_1 \dots f_{i-1} f_i f_{i+1} \dots f_{j-1} f_j f_{j+1} \dots\big) = \sum_{i<j} (-1)^{\cdots} Q_1(f_i f_j)\big(f_1 \dots f_{i-1} f_{i+1} \dots f_{j-1} f_{j+1} \dots\big), \tag{C.2}$$

signs from moving f_i, f_j to the front. Properties stated (p. 25): respects gauge symmetry and the perturbative $U(2)$; turns one b into a c and adds two holomorphic derivatives with contracted spacetime indices.

Pairs, from the diagram computed in [32] (App. C; open adjoint indices; the δ_{AB} structure is that of $SU(2)$, see §8):

$$Q_1(b_A b_B) = \partial_\alpha b_A(x) \partial^\alpha c^B(x) + \partial_\alpha b_B(x) \partial^\alpha c^A(x) - 2\delta_{AB}\, \partial_\alpha b_C(x) \partial^\alpha c^C(x), \tag{C.3}$$

$$Q_1(b_A c^B) = (-1)\Big(\partial_\alpha c^A(x) \partial^\alpha c^B(x) - \delta_A^B\, \partial_\alpha c_D(x) \partial^\alpha c^D(x)\Big). \tag{C.4}$$

$\operatorname{Tr} b^2$ (p. 25):

$$Q_1 \operatorname{Tr} b^2 \propto \partial_\alpha \big(\operatorname{Tr} b\, \partial^\alpha c \big) = \partial_\alpha S^\alpha. \tag{3.40}$$

Claims: $\partial_\alpha S^\alpha$, non-trivial at tree level, is Q-exact at one loop; by (3.43) all higher Q_n act trivially on $\operatorname{Tr} b^2$, so (3.40) receives no further perturbative correction. No coefficient is printed in (3.40).

Planar, single-trace (§5.2). Normalisation: N^{-1} per propagator, N per interaction vertex, single traces with an overall power of N (not specified); the planar differential maps single-trace to single-trace with no N; $1 \to 2$ traces suppressed by N^2, $2 \to 1$ of order 1.

$$Q_1 A_n \sim n \operatorname{Tr} b^{n-2} \partial_\alpha b\, \partial^\alpha c, \tag{5.20}$$

$$Q_1 A_n = \frac{2n}{n-1}\, \partial_\alpha B^\alpha_{n-1}. \tag{5.21}$$

(5.21) after reordering letters with tree-exact terms such as $Q_0 \operatorname{Tr} b^{k_1} \partial_\alpha b\, b^{k_2} \partial^\alpha b$, i.e. on Q_0-cohomology. Further statements (p. 42–43): $Q_1 B_{n,\alpha}$ must be a multiple of $\partial_\alpha C_n$ up to tree-exact terms unless an accidental cancellation occurs (coefficient not given); $Q_1 C_n = 0$ (no single-trace tree classes with three c); one-loop cohomology: $U(N)$ keeps C_n (no derivatives) and derivatives of $\operatorname{Tr} c$; $SU(N)$ keeps only C_n; derivatives of C_n trivial; footnote 42: the $2 \to 1$ part does not change this; quantum numbers forbid further perturbative corrections; even ghost number forbids non-perturbative ones. Index check: $x \to pq$ once loop corrections are included (p. 49, (7.7)).

Bootstrap of Q_1 for $SU(2)$ (App. C). Ansatz

$$Q_1(b_1 c_1) = C_2\, \partial_1 c_1 \partial_2 c_1 + C_1\, \partial_1 c_2 \partial_2 c_2 + \dots \tag{C.5}$$

(adjoint indices); comparison with (C.4) stated to give $C_2 = 0$, $C_1 = -2$. Up to three loops, (3.41) has a unique solution parametrised by $C_{1,2}$:

$$Q_1\big(\partial_1^{n_1}\partial_2^{n_2} b_A\, \partial_1^{m_1}\partial_2^{m_2} c_B\big) = -C_2\, \mathcal{F}(c_A, c_B) + C_1 \frac{2\epsilon_{ADE}\epsilon_{BCE}}{m_1+m_2+2}\, \mathcal{G}(c_C, c_D), \tag{C.6}$$

$$Q_1\big(\partial_1^{n_1}\partial_2^{n_2} b_A\, \partial_1^{m_1}\partial_2^{m_2} b_B\big) = -C_2\, \mathcal{H}(b_A, b_B, c_A, c_B) + C_1 \frac{2\epsilon_{ADE}\epsilon_{BCE}}{m_1+m_2+2}\, \mathcal{K}(b_C, b_D, c_C, c_D), \tag{C.7}$$

$$\begin{aligned}
\mathcal{F}(c_A, c_B) &= \partial_1^{n_1+1}\partial_2^{n_2} c_A\, \partial_1^{m_1}\partial_2^{m_2+1} c_B - \partial_1^{n_1}\partial_2^{n_2+1} c_A\, \partial_1^{m_1+1}\partial_2^{m_2} c_B, \\
\mathcal{H}(b_A, b_B, c_A, c_B) &= \partial_1^{n_1+1}\partial_2^{n_2} c_A\, \partial_1^{m_1}\partial_2^{m_2+1} b_B - \partial_1^{n_1}\partial_2^{n_2+1} c_A\, \partial_1^{m_1+1}\partial_2^{m_2} b_B \\
&\quad - \partial_1^{n_1+1}\partial_2^{n_2} b_A\, \partial_1^{m_1}\partial_2^{m_2+1} c_B + \partial_1^{n_1}\partial_2^{n_2+1} b_A\, \partial_1^{m_1+1}\partial_2^{m_2} c_B, \\
\mathcal{G}(c_C, c_D) &= \sum_{k=0}^{m_1}\sum_{l=0}^{m_2} \frac{1}{k+l+1}\binom{m_1}{k}\binom{m_2}{l}\Big[\partial_1^{m_1-k+1}\partial_2^{m_2-l} c_C\, \partial_1^{n_1+k}\partial_2^{n_2+l+1} c_D \\
&\quad - \partial_1^{m_1-k}\partial_2^{m_2-l+1} c_C\, \partial_1^{n_1+k+1}\partial_2^{n_2+l} c_D\Big], \\
\mathcal{K}(b_C, b_D, c_C, c_D) &= \sum_{k=0}^{m_1}\sum_{l=0}^{m_2} \frac{1}{k+l+1}\binom{m_1}{k}\binom{m_2}{l}\Big[\partial_1^{m_1-k+1}\partial_2^{m_2-l} c_C\, \partial_1^{n_1+k}\partial_2^{n_2+l+1} b_D \\
&\quad - \partial_1^{m_1-k}\partial_2^{m_2-l+1} c_C\, \partial_1^{n_1+k+1}\partial_2^{n_2+l} b_D \\
&\quad - \partial_1^{m_1-k+1}\partial_2^{m_2-l} b_C\, \partial_1^{n_1+k}\partial_2^{n_2+l+1} c_D + \partial_1^{m_1-k}\partial_2^{m_2-l+1} b_C\, \partial_1^{n_1+k+1}\partial_2^{n_2+l} c_D\Big].
\end{aligned}$$

Q_1 so obtained is not nilpotent; two-loop examples from $Q_1^2 + \{Q_0, Q_2\} = 0$:

$$Q_2\big(b_A b^A b_B b^B\big) = \frac{20 C_1^2}{3} \epsilon_{ABC}\, b^A \partial_\alpha \partial_\beta b^B \partial^\alpha \partial^\beta c^C, \tag{C.8}$$

$$Q_2\big(b_A b^A \partial_\alpha c_B \partial^\alpha c^B\big) = -\frac{10 C_1^2}{9} \epsilon_{ABC}\, \partial_\alpha c^A \partial^\alpha \Big[\partial_1^2 c^B \partial_2^2 c^C - \partial_1\partial_2 c^B \partial_1\partial_2 c^C\Big]. \tag{C.9}$$

Also (p. 26): the general structure plus $\{Q_0, Q_1\} = 0$ bootstraps Q_1 up to two coefficients fixed by direct integration; the analogous statement appears to hold for all Q_n.

Numerics (§8, $SU(2)$ up to $M_R = 14$, $SU(3)$ up to $M_R = 8$; (8.1)). Steps: basis of $\mathcal{V}_{|b|}(n_1, n_2)$; Q_0 cohomology; then

$$Q_1 : H^i(\mathcal{V}_{|b|}(n_1, n_2), Q_0) \to H^{i+1}(\mathcal{V}_{|b|-1}(n_1, n_2), Q_0), \tag{8.3}$$

$$Q_2 : H^\bullet(H^\bullet(\mathcal{V}_{|b|}(n_1, n_2), Q_0), Q_1) \to H^\bullet(H^\bullet(\mathcal{V}_{|b|-2}(n_1, n_2), Q_0), Q_1), \tag{8.4}$$

$$H^\bullet(\mathcal{V}_\bullet(n_1, n_2), Q_{\text{pert}}) \equiv \bigoplus_{|b|} H^{\bullet+n_1+n_2}\big(H^\bullet(\mathcal{V}_{|b|}(n_1, n_2), Q_0), Q_1\big). \tag{8.5}$$

Within the range, the one-loop cohomology is sparse enough to forbid higher-loop differentials, so (8.5) is claimed exact to all orders. One-loop representatives (Table 3): $SU(2)$: $M_R = 2$: C_0; 6: C_2; 10: C_4, $B_{1,2}\partial_2 C_0 \partial_1^2 C_0$; 12: $B_{1,2}\partial_1\partial_2 B_{1,2}\partial_1^2 C_0$; 14: C_6, $B_{3,2}\partial_2 C_0\partial_1^2 C_0$; none at $4, 8$. $SU(3)$: $M_R = 2$: C_0; 4: C_1, C_0^2. $C_0 = \partial_\alpha c\, \partial^\alpha c$ (trace understood) = gaugino bilinear (p. 4, 52).

6 Konishi anomaly and the stress tensor

$$(Q + \bar\partial) S_\alpha \sim \partial_\alpha \operatorname{Tr}\big(\partial_\beta c\, \partial^\beta c\big), \tag{3.24}$$

$$\int \mathrm{d}^2 z\, v^\alpha \partial_\alpha \operatorname{Tr}\big(\partial_\beta c\, \partial^\beta c\big), \tag{3.25}$$

$$\operatorname{Tr}([c, b] \circ \partial_\alpha c) + \operatorname{Tr}(b \circ [c, \partial_\alpha c]). \tag{3.26}$$

p. 20: pure gauge theory is not scale invariant quantum mechanically; scale invariance and $U(1)_R$ broken at one loop; the R-anomaly ($\propto \operatorname{Tr} F \wedge F$, triangle, one-loop exact) is supersymmetrised to the Konishi anomaly equation involving $\operatorname{Tr} \overline{\mathcal{W}}_\alpha \overline{\mathcal{W}}^\alpha$ [28,30,31]; (3.24) is obtained by translating [53] (Yonekura), to be recovered by a one-loop triangle in BF theory. (3.25): an infinitesimal holomorphic coordinate change v^α shifts the action by an amount $\propto$ (3.25), i.e. shifts τ by a multiple of $\partial_\alpha v^\alpha$ (holomorphic generalisation of the scale anomaly); divergence-free v: no anomaly, complex symplectomorphisms survive. Footnote 28: Q obeys Leibniz only for the loop-corrected product $\mathcal{O} \circ \mathcal{O}' \equiv \widehat{\mathcal{O}}_{-1,-1}\mathcal{O}'$; this product is not associative; (3.26) fails to vanish by a one-loop correction to the regularised product and reproduces the anomaly. Footnote 41: the same non-associativity obstructs absorbing trace-number mixing.

§2.4 (p. 12): $S_\alpha \equiv D_{\dot+} S_{\dot+\alpha}|_{\theta=0}$ (2.20) from the $\mathcal{S}$-multiplet [48]; $(Q + \bar\partial) S_\alpha = \bar Y_{\dot+\alpha}$, $\partial_{\dot+\alpha} \bar Y_{\dot+}^\alpha = 0$ (2.21); $\partial_{z^\alpha} S^\alpha$ is semi-chiral; $\bar Y$ vanishes with a conserved R-current. For a classically scale-invariant theory with one-loop beta function: $(Q + \bar\partial) S_\alpha$ vanishes classically but not at one loop, $(Q + \bar\partial)\partial_\alpha S^\alpha$ vanishes exactly. Modes of $\partial_\alpha S^\alpha$ generate Hamiltonian holomorphic symplectomorphisms:

$$\int_{S^3} f(z) \partial_\alpha S^\alpha(z) \mathcal{O}(w) = -\int_{S^3} (\partial_\alpha f(z)) S^\alpha(z) \mathcal{O}(w). \tag{2.22}$$

Holomorphic confinement (p. 12, 26): $\partial_\alpha S^\alpha$ is $(Q + \bar\partial)$-exact at one loop in pure gauge theory; since its modes include translations (B.11), holomorphic derivatives of all local operators vanish in cohomology; the twisted theory is topological at one loop (expected to persist to all orders and non-perturbatively). Index: $I_{\text{pert}}^{SU(2)}[p, q] = \sum_{\ell \geq 0} (pq)^{\ell(\ell+1)/2}$ (7.6), a function of pq only, consistent with this.

Introduction (p. 3): the one-loop corrections are semi-chiral analogues of the generalised Konishi anomaly equations [28,30,31]; corrections cannot stop at one loop since the one-loop differential is not nilpotent.

7 Regulator, scheme, renormalisation freedom

Item	Statement (location)
regulator for Q_0	interaction removed in a ball $\|z\| < \epsilon$ around $\mathcal{O}$ (3.30); result ϵ-independent in Q_{free}-cohomology (p. 22); $\epsilon \to 0$ projects to $\mathcal{V}$ (3.32).
regulator for $Q_{n\geq 1}$	regularised propagators (p. 23), not specified here; final integrals claimed finite and regulator independent (step 5, p. 24). Explicit triangle evaluation in [32]; scheme in [26].
scheme dependence	**Q** is scheme dependent; **Q**'s of different schemes are expected to be related by quasi-isomorphisms (p. 22). Parametrisation of the deformation space is scheme dependent beyond first order (App. E, p. 64).
products	Leibniz rule only for the loop-corrected product; regularised product not associative (fn. 28, 41).
Q_0-exact ambiguity	(3.39): $Q_0 b = [c, b]$ up to a Q_{free}-exact term; (5.21) stated after adding tree-exact terms; $Q_1 B_{n,\alpha} \propto \partial_\alpha C_n$ up to tree-exact terms; (B.13) up to Q_0-exact terms.
scheme-independent content	cohomology (quasi-isomorphism class) of **Q**; within the filtration (3.43), $Q_1 \operatorname{Tr} b^2 \propto \partial_\alpha S^\alpha$ exact to all perturbative orders (p. 25).
free theory	composites of $b^{(0)}, c^{(0)}$: no renormalisation ambiguity (p. 15–16).
choice of contour	footnote 4: non-perturbatively the path-integration contour / solution of Ward identities might not be unique.
counterterms, field redefinitions	no explicit statement on how a finite counterterm or a field redefinition changes Q_1; not discussed in this paper.

8 Other generators at one loop

- $U(2)$ ($= SU(2)$ of $\bar{M}_{\alpha\beta}$ and $M_R = M_{+\dot{-}} - R$): $U(1)_R$ anomalous at one loop in the theories considered; $U(2)$ formally survives in perturbation theory, broken to $SU(2)$ by instantons (p. 6); the perturbative differential commutes with $U(2)$ (p. 51).
- Translations and holomorphic symplectomorphisms: generated by modes of $\partial_\alpha S^\alpha$ ((2.22), (B.8)–(B.11)); footnote 27: translations of a classically conserved current can receive quantum corrections ($P_\mu J^\mu = \partial_\mu J^\mu$).
- Scale/holomorphic vector fields: (3.25).
- S, K, Δ (superconformal): only the algebraic relations (2.4), (2.5); pure $\mathcal{N} = 1$ SYM is not conformal; no one-loop statement on the action of special supercharges, special conformal generators or dilatation on operators. No $\mathcal{N} = 4$ one-loop statement.
- App. F.2 (p. 72–73): "twist" = passing to Q-cohomology; twisting deforms Q_{BRST} by Q, with a $U(1) \subset$ R-symmetry used to regrade; twisted stress tensor $T' = \{Q, S\}$ (S here an operator, not a special supercharge).

9 Our checks on App. C

Script `bgkwwy_checks.py`, output `bgkwwy_checks_output.txt` (this folder). Not statements of the paper. $SU(2)$, $f^{ABC} = \epsilon^{ABC}$, (3.20), (3.23). Two readings of $X_\alpha Y^\alpha$: (K1) $X_1Y_2 - X_2Y_1$ (footnote 54), (K2) $X_2Y_1 - X_1Y_2$.

1. (C.3) $= -(f_{ACE}f_{BDE}+f_{BCE}f_{ADE})\,\partial_\alpha b^C \partial^\alpha c^D$ and (C.4) $= f_{ACE}f_{BDE}\,\partial_\alpha c^C \partial^\alpha c^D$ (both readings). Extension of these forms to general G is not in the paper.

2. From (C.3), (3.20), (3.23): $Q_1 \operatorname{Tr} b^2 = -4\,\partial_\alpha \operatorname{Tr}(b\,\partial^\alpha c)$ (both readings), i.e. the coefficient of (3.40) with S_α as in (3.19) is -4 (-2 with the App. B.1 normalisation $2\operatorname{Tr} b\partial_\alpha c$). (5.21) at $n = 2$ gives $+4$ in the planar normalisation of §5.2; the two normalisations are not related explicitly in the paper.

3. (C.4) at $A = B = 1$ gives $C_1 = +2$ in (C.5) with (K1), $C_1 = -2$ with (K2); the paper states $C_1 = -2$.

4. (C.6), (C.7) at $n_i = m_i = 0$ with $(C_1, C_2) = (-2, 0)$ equal $-2\times$(C.4), $-2\times$(C.3) (K1) or $+2\times$ (K2). They reproduce (C.3), (C.4) iff $C_2 = 0$ and $C_1 = +1$ (K1) or $C_1 = -1$ (K2), while (C.5) read off from (C.4) gives $C_1 = +2$ (K1) or -2 (K2): in both readings the C_1 of (C.6)–(C.7) is one half of the C_1 of (C.5); with the printed value $C_1 = -2$ inserted in (C.6)–(C.7) all their outputs are twice the diagrammatic values (K2).

5. With (C.6) so normalised: $Q_1 S_\alpha = \partial_\alpha \operatorname{Tr}(\partial_\beta c\,\partial^\beta c)$, $\alpha = 1, 2$, both readings: the coefficient replacing $\sim$ in (3.24) is 1 for $SU(2)$ in this normalisation. Consistency: $Q_1(Q_1 \operatorname{Tr} b^2) = -4\,\partial_\alpha\partial^\alpha \operatorname{Tr}(\partial_\beta c\,\partial^\beta c) = 0$, as required by (3.42) at $n = 2$ on $\operatorname{Tr} b^2$ (where $Q_2 \operatorname{Tr} b^2 = 0$ by (3.43)).

10 References of the paper relevant here

[19] Chang–Yin, arXiv:1305.6314 (tree-level $\mathcal{N} = 4$ superfield (4.4), large-N conjecture). [26] Gaiotto–Kulp–Wu, *Higher operations in perturbation theory*, arXiv:2403.13049 (scheme, all-loop corrections to Q). [28] Cachazo–Douglas–Seiberg–Witten, hep-th/0211170. [30] Konishi, PLB 135 (1984) 439. [31] Konishi–Shizuya, Nuovo Cim. A 90 (1985) 111. [32] Budzik–Gaiotto–Kulp–Wu–Yu, *Feynman diagrams in four-dimensional holomorphic theories and the Operatope*, arXiv:2207.14321 (triangle for (C.3)–(C.4); two-loop diagram). [51] Chang–Lin, arXiv:2209.06728 ($\mathcal{N} = 4$ $SU(N)$ numerics, non-multitrace class at $M_R \simeq 8$ for $SU(2)$, p. 53). [53] Yonekura, arXiv:1004.1296 (source of (3.24)).